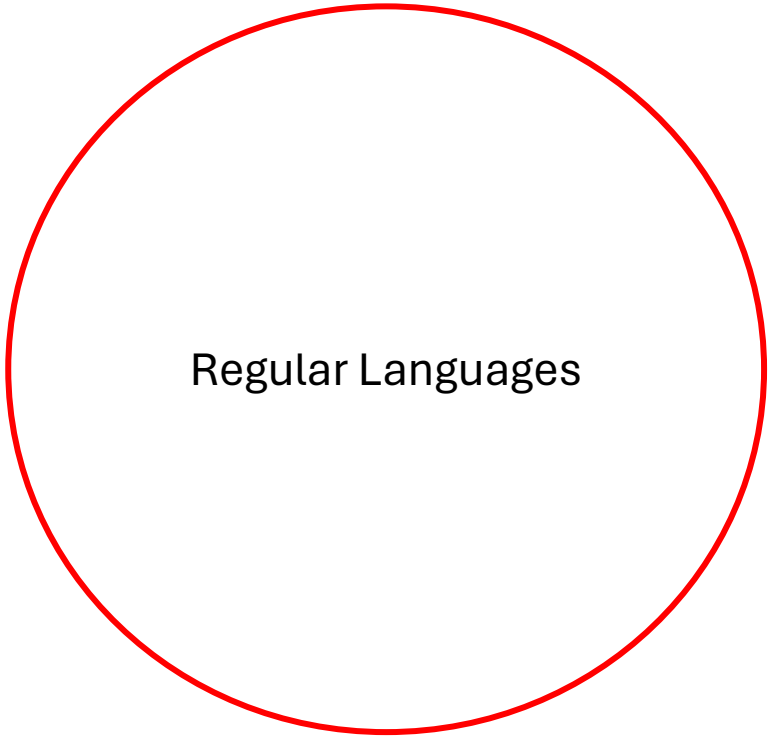
The Theory of Computing

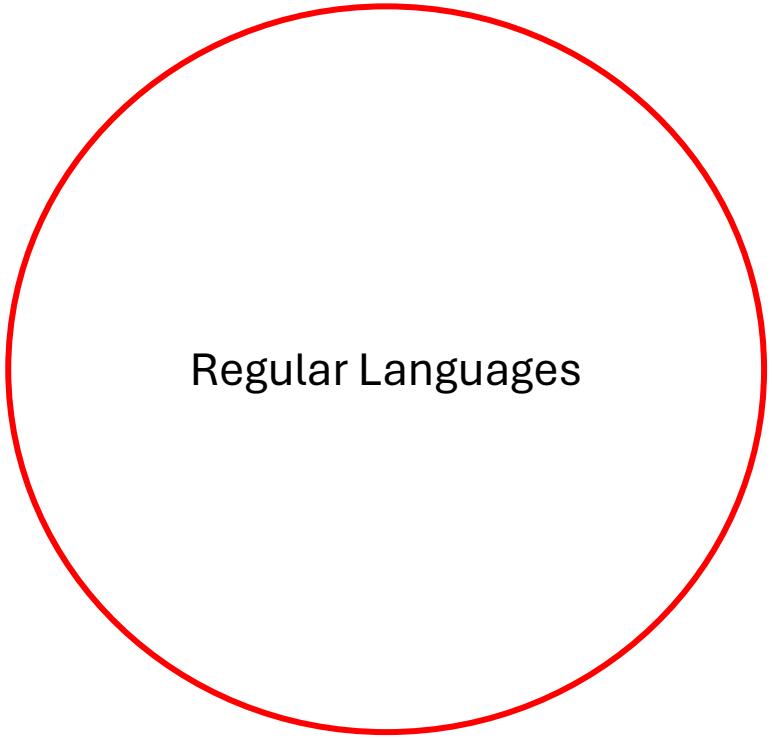
Non-Regular Languages

- L



There are languages outside the regular languages.

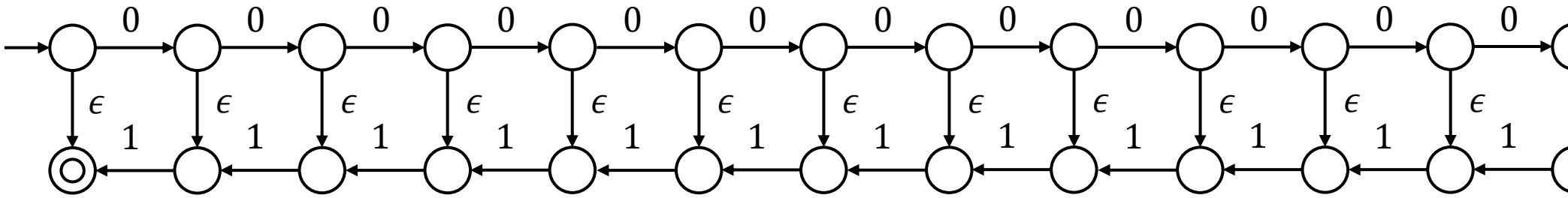
- L



For instance, $L = \{0^n 1^n \mid n \geq 0\}$.

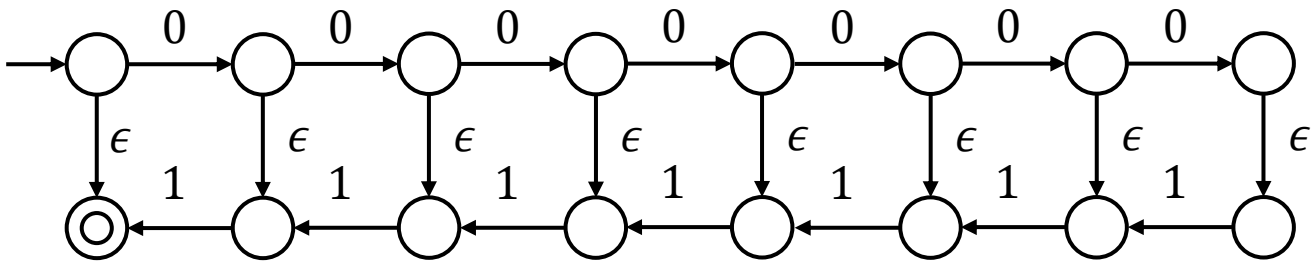
$$L = \{0^n 1^n \mid n \geq 0\}$$

To make an NFA for this you would need infinite states.



$$L = \{0^n 1^n \mid 0 \leq n \leq 7\}$$

If we place a finite upper limit, L is regular.



$$L = \{0^n 1^n \mid n \geq 0\}$$

You would also need an infinite regular expression.

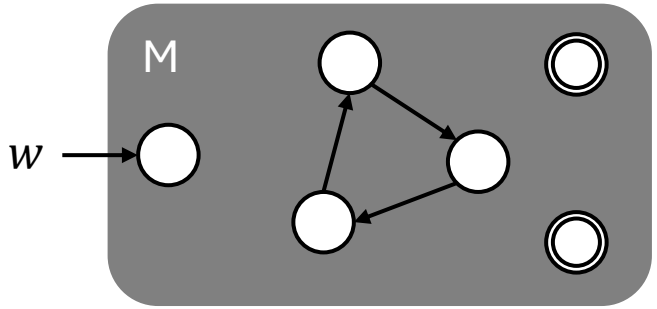
$$\epsilon \cup 01 \cup 0011 \cup 000111 \cup 00001111 \cup \cdots$$

$$L = \{0^n 1^n \mid n \geq 0\}$$

These arguments do not constitute proofs that this language is not regular.

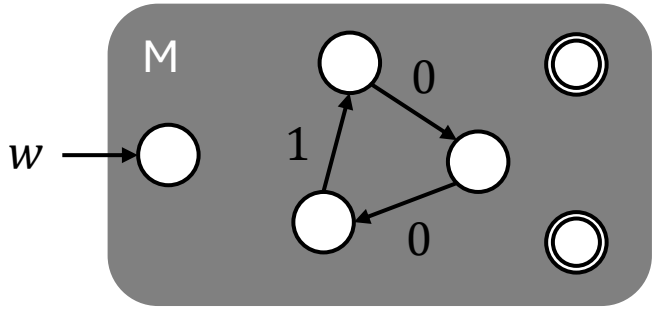
To prove languages are not regular we use a theorem called the pumping lemma.

Consider a regular language with infinite strings.



A DFA (or NFA) for a regular language with infinite strings must have a loop.

An RE for a regular language with infinite strings must have a star.



The loop or star will allow a pattern to be repeated (pumped) in strings of regular languages.

Pumping Lemma: If L is a regular language, then there exists a number $p \geq 0$ (the pumping length) such that if $w \in L$ and $|w| \geq p$ then w can be divided into three pieces $w = xyz$ satisfying these three conditions:

1. For each $i \geq 0$, $xy^i z \in L$.
2. $|y| > 0$
3. $|xy| \leq p$

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1. For each $i \geq 0$, $xy^iz \in L$.
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The substring y represents the part that can be repeated (pumped) by following the loop. This loop can be taken 0 or more times to produce other strings in the language.

Pumping Lemma: If L is a regular language, then there exists a number $p \geq 0$ (the pumping length) such that if $w \in L$ and $|w| \geq p$ then w can be divided into three pieces $w = xyz$ satisfying these three conditions:

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The condition that $|xy| \leq p$ states that the repeated pattern must be detected after a finite number of states.

Pumping Lemma: If L is a regular language, then there exists a number $p \geq 0$ (the pumping length) such that if $w \in L$ and $|w| \geq p$ then w can be divided into three pieces $w = xyz$ satisfying these three conditions:

1. For each $i \geq 0$, $xy^i z \in L$.
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Proof Strategy: If we can show that a language does not have this property, then we can show that it is not regular.

$$L = \{0^n 1^n \mid n \geq 0\}$$

We can use the pumping lemma to show that L is not regular.

Theorem: $L = \{0^n 1^n \mid n \geq 0\}$ is not regular.

Proof: Assume that L is a regular language so the pumping lemma for regular languages must hold.

(We will seek a contradiction to show that this assumption is false.)

Theorem: $L = \{0^n 1^n \mid n \geq 0\}$ is not regular.

Proof (continued): If the pumping lemma holds, then any string longer than the pumping length p will have the pumping properties.

For instance, the string $s = 0^p 1^p$ is, by design, longer than the pumping length p, so s will have the pumping properties.

Theorem: $L = \{0^n 1^n \mid n \geq 0\}$ is not regular.

Proof (continued): By the pumping properties $s = 0^p 1^p$ can be split into three parts $s = xyz$ such that:

1. $|xy| \leq p$
2. $|y| > 0$
3. For each $i \geq 0$, $xy^i z \in L$.

Theorem: $L = \{0^n 1^n \mid n \geq 0\}$ is not regular.

Proof (continued): By the pumping properties $s = 0^p 1^p$ can be split into three parts $s = xyz$ such that:

1. $|xy| \leq p$ means that both x and y must be in 0^p.
2. $|y| > 0$
3. For each $i \geq 0$, $xy^i z \in L$.

Theorem: $L = \{0^n 1^n \mid n \geq 0\}$ is not regular.

Proof (continued): By the pumping properties $s = 0^p 1^p$ can be split into three parts $s = xyz$ such that:

1. $|xy| \leq p$ means that both x and y must be in 0^p.
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Proof (continued): By the pumping properties $s = 0^p 1^p$ can be split into three parts $s = xyz$ such that:

1. $|xy| \leq p$ means that both x and y must be in 0^p.
2. $|y| > 0$ means that y must have at least one 0.
3. For each $i \geq 0$, $xy^i z \in L$.

Let's pick $i = 0$, then $xz = 0^{p-|y|}1^p$. Since $|y| > 0$, $xz \notin L$ and this property doesn't hold.

Theorem: $L = \{0^n 1^n \mid n \geq 0\}$ is not regular.

Proof (continued): By the pumping properties $s = 0^p 1^p$ can be split into three parts $s = xyz$ such that:

1. $|xy| \leq p$ means that both x and y must be in 0^p.
2. $|y| > 0$ means that y must have at least one 0.
3. For each $i \geq 0$, $xy^i z \in L$.

Let's pick $i = 2$, then $xy^2z = 0^{p+|y|}1^p$. Since $|y| > 0$, $xy^2z \notin L$ and this property doesn't hold.

Theorem: $L = \{0^n 1^n \mid n \geq 0\}$ is not regular.

Proof (continued): Since the pumping property did not hold for our selected string s, we've shown that the pumping lemma doesn't hold for L.

This contradicts our assumption that L was a regular language. So, L must be non-regular. □

Theorem: $L = \{0^n 1^n \mid n \geq 0\}$ is not regular.

Proof: Assume that L is a regular language so the pumping lemma for regular languages must hold.

The string $s = 0^p 1^p$ is by design longer than the pumping length p, so according to the pumping lemma this string will have the pumping properties.

By the pumping properties $s = 0^p 1^p$ can be split into three parts $s = xyz$ such that:

1. $|xy| \leq p$ meaning that both x and y must be in 0^p.
2. $|y| > 0$ meaning that y must have at least one 0.
3. For each $i \geq 0$, $xy^i z \in L$.

Let's pick $i = 0$, then $xz = 0^{p-|y|}1^p$. Since $|y| > 0$, $xz \notin L$ and this property doesn't hold.

This contradicts our assumption that L was a regular language. So, L must be non-regular. □

$$L = \{w \in \{0,1\}^* \mid w \text{ has an equal number of } 0s \text{ and } 1s \}$$

We can use the closure properties of regular languages to show that L is not regular.

$$L = \{w \in \{0,1\}^* \mid w \text{ has an equal number of } 0s \text{ and } 1s\}$$

Proof: The language $L_R = \{0^*1^*\}$ is regular.

If we take $L \cap L_R$ we get the language $L_N = \{0^n 1^n \mid n \geq 0\}$.

We have already shown that L_N is not a regular language.

If L was regular, then since the regular languages are closed under intersection, $L_N = L \cap L_R$ should be regular.

Since we have shown that L_N is not regular, we can conclude that L is not regular either. □

Exercises

$L_0 = \{a^n b^n c^n \mid w \in \{a, b, c\}^*\}$

$L_1 = \{ww \mid w \in \{0,1\}^*\}$

$L_2 = \{0^n 1^m \mid w \in \{0,1\}^* \text{ and } n \leq m\}$

$L_3 = \{0^n 1^m \mid w \in \{0,1\}^* \text{ and } n \neq m\}$

$L_4 = \{w \in \{0,1\}^* \mid w \text{ is a palindrome}\}$